

ACME Corporation

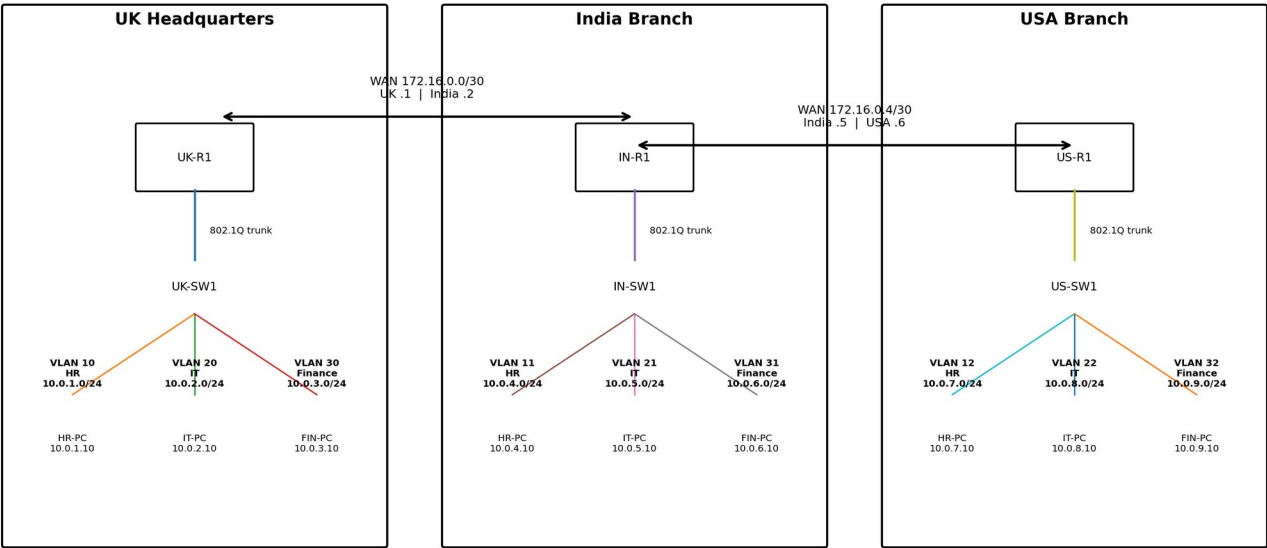
Secure Network Design Portfolio Case Study

Corrected portfolio reconstruction based on completed university work

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ACME Corporation - Corrected Logical Network Topology

Three sites, departmental VLANs, router-on-a-stick and RIPv2 route exchange



Generated diagrams and terminal-style images are explanatory portfolio assets, not authentic Packet Tracer captures.

1. Executive Summary

This portfolio reconstruction presents a corrected and internally consistent network design for ACME Corporation. The network connects offices in the UK, India and USA while separating HR, IT and Finance with VLANs.

Technical explanation: The design uses nine /24 departmental subnets, two /30 WAN links, router-on-a-stick, RIPv2, port security and basic device hardening.

Simple explanation: Each country has its own office network. Departments are placed into separate sections, and routers share directions so devices can communicate between countries.

2. Original Project and Reconstruction

The original coursework contained strong topics but mixed multiple IP-address schemes and generic screenshots. This version keeps the intended networking concepts while correcting the design and clearly separating generated illustrations from authentic implementation evidence.

- Standardised the design around 10.0.0.0/20 and 172.16.0.0/30 WAN links.
- Defined router-on-a-stick as the inter-VLAN routing method.
- Corrected the RIPv2 network statement.
- Removed plaintext example credentials.
- Generated new diagrams and sanitised configuration templates.

3. Corrected Network Topology

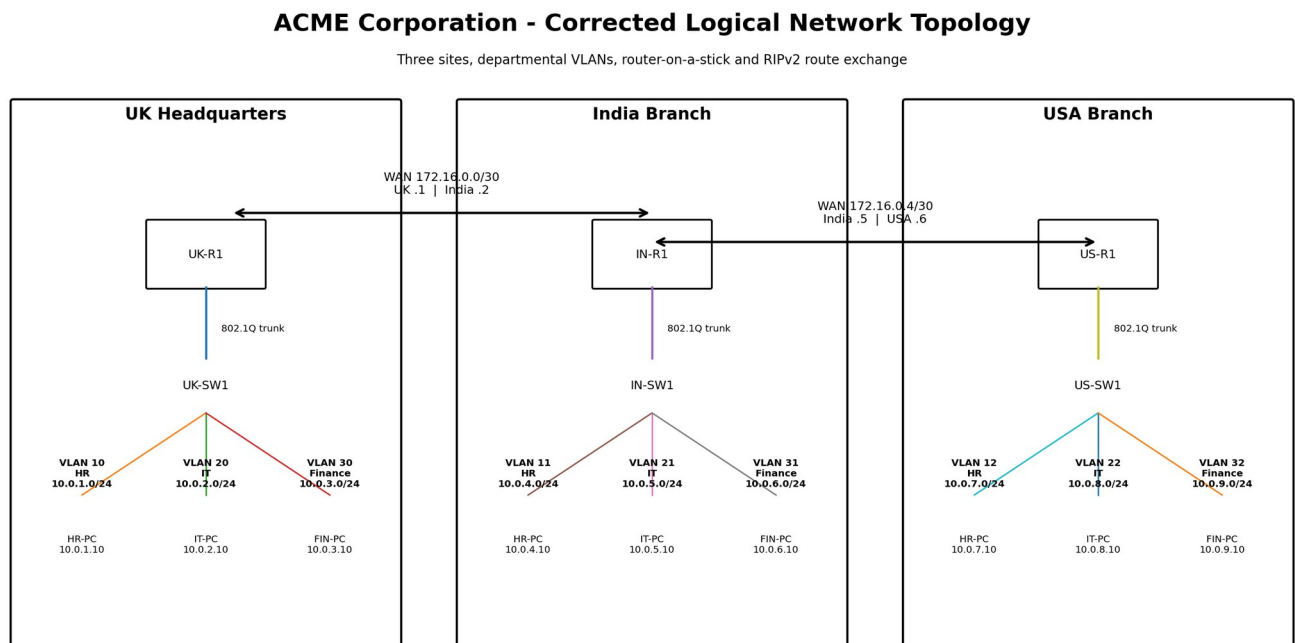


Figure 1. Corrected logical topology for the three ACME sites.

4. Addressing and Subnetting

Technical explanation: Fixed-Length Subnet Masking allocates equal /24 networks to the nine departments. Each /24 provides 254 normally usable host addresses. The /30 WAN networks provide two usable router addresses.

Simple explanation: The company address space is split into predictable blocks. Departments receive large equal blocks, while links between routers receive only the two addresses they need.

Subnet Allocation from 10.0.0.0/20

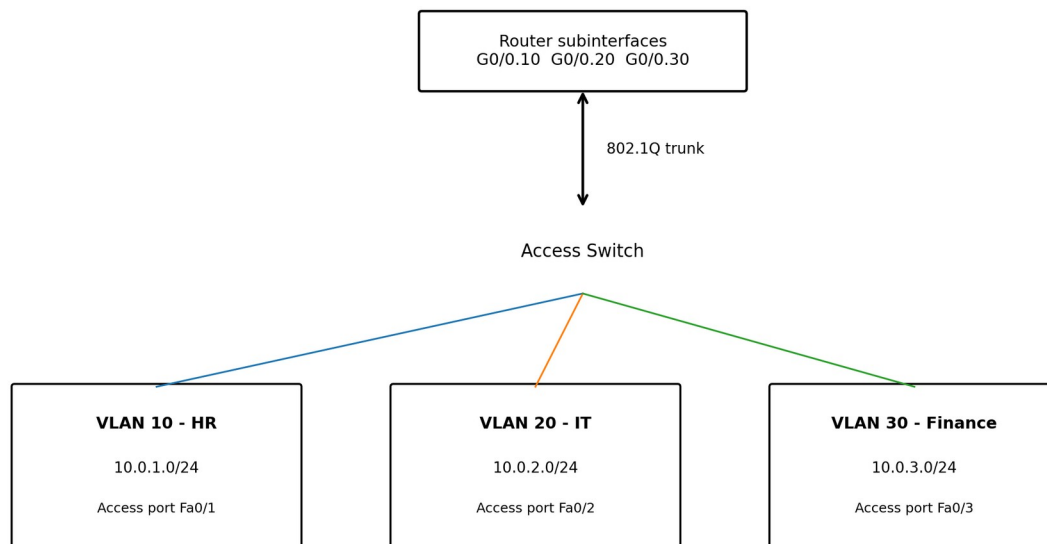
Site	Department	Network	Gateway	Usable host range
UK	HR	10.0.1.0/24	10.0.1.1	10.0.1.1 - 10.0.1.254
UK	IT	10.0.2.0/24	10.0.2.1	10.0.2.1 - 10.0.2.254
UK	Finance	10.0.3.0/24	10.0.3.1	10.0.3.1 - 10.0.3.254
India	HR	10.0.4.0/24	10.0.4.1	10.0.4.1 - 10.0.4.254
India	IT	10.0.5.0/24	10.0.5.1	10.0.5.1 - 10.0.5.254
India	Finance	10.0.6.0/24	10.0.6.1	10.0.6.1 - 10.0.6.254
USA	HR	10.0.7.0/24	10.0.7.1	10.0.7.1 - 10.0.7.254
USA	IT	10.0.8.0/24	10.0.8.1	10.0.8.1 - 10.0.8.254
USA	Finance	10.0.9.0/24	10.0.9.1	10.0.9.1 - 10.0.9.254

WAN links: 172.16.0.0/30 (UK-India) and 172.16.0.4/30 (India-USA)

A /24 allows 254 usable hosts. A /30 is efficient for a two-router point-to-point link.

5. VLANs and Inter-VLAN Routing

How VLAN Separation and Inter-VLAN Routing Work



Technical: VLANs are separate Layer 2 broadcast domains. The router provides Layer 3 gateways between them.

Technical explanation: VLANs form separate Layer 2 broadcast domains. The switch-to-router link is an 802.1Q trunk, and router subinterfaces provide the Layer 3 gateways.

Simple explanation: VLANs create invisible departmental walls. The router provides controlled doorways between them.

6. End-Device Configuration

ILLUSTRATIVE OUTPUT - REPLACE WITH A REAL PACKET TRACER CAPTURE WHEN AVAILABLE

PC-UK-HR - Desktop > IP Configuration

IPv4 Configuration

IPv4 Address

10.0.1.10

Subnet Mask

255.255.255.0

Default Gateway

10.0.1.1

DNS Server

10.0.2.20 (example)

Purpose: assigns the device an address and tells it to use the UK router as its gateway.

This illustrative panel shows the fields required on a UK HR device. It must be replaced with a real Packet Tracer screenshot after the final topology is built.

7. VLAN Verification

ILLUSTRATIVE OUTPUT - REPLACE WITH A REAL PACKET TRACER CAPTURE WHEN AVAILABLE

UK-SW1 - show vlan brief

```
UK-SW1# show vlan brief
```

VLAN	Name	Status	Ports
1	default	active	Fa0/4, Fa0/5, Fa0/6 ...
10	HR	active	Fa0/1
20	IT	active	Fa0/2
30	FINANCE	active	Fa0/3

```
UK-SW1#
```

Generated from the corrected portfolio design; not captured from Cisco Packet Tracer.

The expected output lists the departmental VLANs and assigned access ports. Authentic output should be captured from each site switch.

8. Inter-VLAN Router Configuration

ILLUSTRATIVE OUTPUT - REPLACE WITH A REAL PACKET TRACER CAPTURE WHEN AVAILABLE

UK-R1 - Router-on-a-Stick Configuration

```
UK-R1# show running-config | section GigabitEthernet0/0

interface GigabitEthernet0/0
  no ip address
  no shutdown
!
interface GigabitEthernet0/0.10
  encapsulation dot1Q 10
  ip address 10.0.1.1 255.255.255.0
!
interface GigabitEthernet0/0.20
  encapsulation dot1Q 20
  ip address 10.0.2.1 255.255.255.0
!
interface GigabitEthernet0/0.30
  encapsulation dot1Q 30
  ip address 10.0.3.1 255.255.255.0
```

Generated from the corrected portfolio design; not captured from Cisco Packet Tracer.

Router subinterfaces attach a VLAN tag and gateway address to each departmental network.

9. Port Security

Cisco Switch Port-Security Violation Modes

PROTECT	RESTRICT	SHUTDOWN
Drops unauthorised frames	Drops unauthorised frames	Places the interface into error-disabled state
No alert or counter increase	Records the violation and may alert	Administrator recovery required
Simple: silently ignores the unknown device	Simple: blocks and logs the attempt	Simple: closes the port completely

ILLUSTRATIVE OUTPUT - REPLACE WITH A REAL PACKET TRACER CAPTURE WHEN AVAILABLE

UK-SW1 - show port-security interface fa0/1

```
UK-SW1# show port-security interface fastEthernet 0/1
```

```
Port Security           : Enabled
Port Status             : Secure-up
Violation Mode          : Protect
Maximum MAC Addresses   : 1
Total MAC Addresses     : 1
Configured MAC Addresses : 0
Sticky MAC Addresses    : 1
Security Violation Count : 0
```

```
UK-SW1#
```

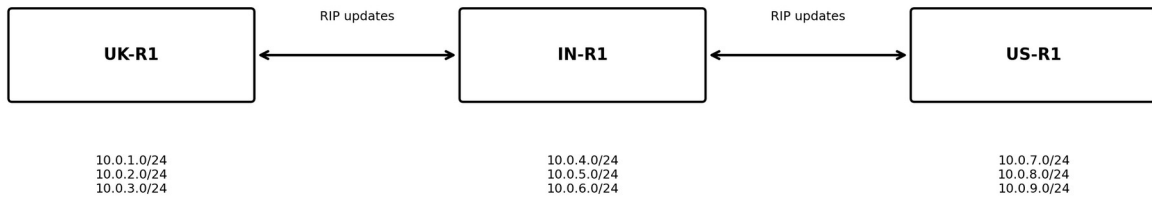
Generated from the corrected portfolio design; not captured from Cisco Packet Tracer.

Technical explanation: sticky MAC learning records the approved source address. Protect, restrict and shutdown apply progressively stronger responses.

Simple explanation: the switch remembers the expected device and reacts when another device appears.

10. Dynamic Routing

RIPv2 Route Exchange



Technical: routers advertise reachable networks and learn remote routes marked 'R' in the routing table.

Simple: each router shares directions to the office networks it can reach.

ILLUSTRATIVE OUTPUT - REPLACE WITH A REAL PACKET TRACER CAPTURE WHEN AVAILABLE

UK-R1 - show ip route

```
UK-R1# show ip route
```

```
C    10.0.1.0/24 is directly connected, GigabitEthernet0/0.10
C    10.0.2.0/24 is directly connected, GigabitEthernet0/0.20
C    10.0.3.0/24 is directly connected, GigabitEthernet0/0.30
C    172.16.0.0/30 is directly connected, Serial0/0/0
R    10.0.4.0/24 [120/1] via 172.16.0.2, Serial0/0/0
R    10.0.5.0/24 [120/1] via 172.16.0.2, Serial0/0/0
R    10.0.6.0/24 [120/1] via 172.16.0.2, Serial0/0/0
R    10.0.7.0/24 [120/2] via 172.16.0.2, Serial0/0/0
R    10.0.8.0/24 [120/2] via 172.16.0.2, Serial0/0/0
R    10.0.9.0/24 [120/2] via 172.16.0.2, Serial0/0/0
```

```
UK-R1#
```

Generated from the corrected portfolio design; not captured from Cisco Packet Tracer.

RIPv2 allows routers to learn remote office networks. The routing table marks RIP-learned routes with R.

11. Connectivity Testing

ILLUSTRATIVE OUTPUT - REPLACE WITH A REAL PACKET TRACER CAPTURE WHEN AVAILABLE

PC-UK-HR - Command Prompt

```
C:\> ping 10.0.7.10

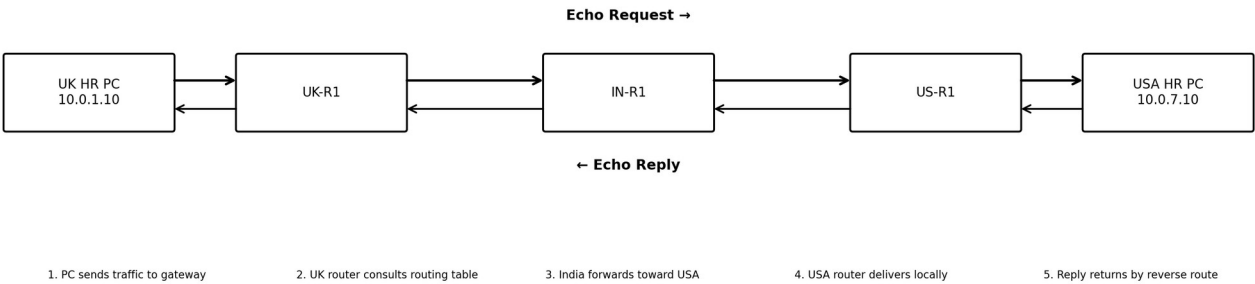
Pinging 10.0.7.10 with 32 bytes of data:
Reply from 10.0.7.10: bytes=32 time<1ms TTL=125
Reply from 10.0.7.10: bytes=32 time<1ms TTL=125
Reply from 10.0.7.10: bytes=32 time<1ms TTL=125
Reply from 10.0.7.10: bytes=32 time<1ms TTL=125

Ping statistics for 10.0.7.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss)

C:\>
```

Generated from the corrected portfolio design; not captured from Cisco Packet Tracer.

ICMP Test Path: UK HR PC to USA HR PC

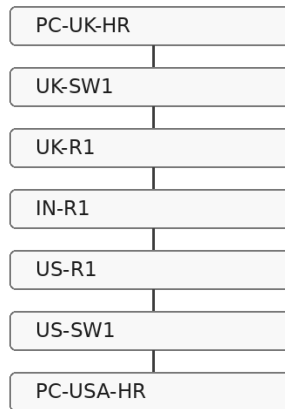


Packet Flow Inspection - ICMP Echo Request

OSI Model / Outbound PDU Details

Source device	PC - UK - HR
Source IP	10.0.1.10
Destination IP	10.0.7.10
Protocol	ICMP
Type	8 - Echo Request
Next hop	10.0.1.1

Expected path

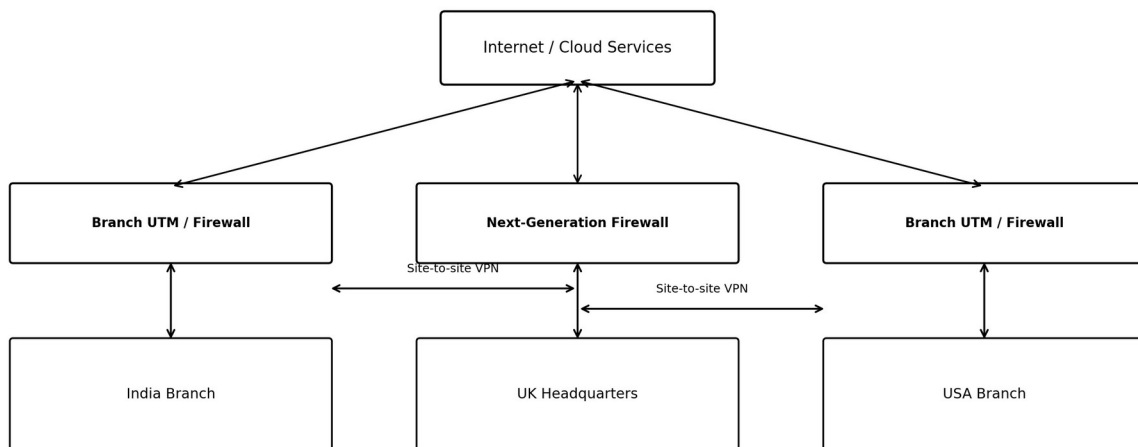


Generated from the corrected design; this is an explanatory mock-up, not a captured Packet Tracer event window.

Ping confirms bidirectional Layer 3 reachability. Simulation-style inspection explains how the ICMP packet moves through the routers.

12. Security and Firewall Recommendations

Recommended Firewall and VPN Placement



Technical: firewalls enforce policy at site boundaries; encrypted VPN tunnels protect inter-site traffic.

A realistic deployment would place a next-generation firewall at headquarters, smaller branch firewalls at India and USA, and encrypted site-to-site VPNs between locations.

- Use SSH rather than Telnet.

- Add access-control lists between departments.
- Add a dedicated management VLAN.
- Use central logging and SNMPv3.
- Shut down unused switch ports.
- Introduce DHCP snooping, Dynamic ARP Inspection and BPDU Guard.

13. Limitations

- The generated images are explanatory and are not authentic Packet Tracer screenshots.
- The reference configurations must be tested against the actual Cisco models and interface names used.
- RIP is suitable for a small teaching topology but is not the preferred protocol for a growing enterprise.
- VLANs provide segmentation but do not replace ACLs or firewalls.

14. Future Improvements

- Rebuild the topology in Packet Tracer and replace illustrative outputs with authentic screenshots.
- Replace RIP with OSPF.
- Add ACLs, firewalls and VPNs.
- Add redundancy and failure testing.
- Add central monitoring and configuration backups.

15. Skills Demonstrated

- IPv4 subnetting and addressing
- VLAN configuration and trunking
- Router-on-a-stick
- RIPv2 routing
- Switch port security
- Cisco IOS configuration
- Network testing and troubleshooting
- Security design and documentation

Appendix A. Reference Configuration Files

Sanitised router and switch templates are included in the GitHub repository under configurations/. They contain redacted credential placeholders and must be verified in Packet Tracer.